PRN = 202501080014

NAME = YASH MILIND CHAUDHARI

BATCH & DIVISION = IT1 & I(11)

PRACTICAL NO.1

Practice Lab Assignments:

1. To accept an object mass in kilograms and velocity in meters per second and display its momentum.

Momentum is calculated as e=mc2 where m is the mass of the object and c is its velocity

Ans :
```
m = float(input("Enter mass(kg): "))
v = float(input("Enter velocity(m/s): "))
momentum = m * v
print("Momentum = ",momentum, "kg.m/s")
```

2. Write a Python program for following conditions.

- If n is single digit print square of it.
- If n is two digit print square root of it.
- If n is three digit print cube root of it.

Ans :
```
import math
n = int(input("Enter a number: "))
if 0 < n < 10:
    square = n * n
```

```
print("Square = ", square)
elif 10 < n < 100:
    square_root = n ** 0.5
    print("Square root = ", square_root)
elif 100 < n < 1000:
    cube_root = n ** (1/3)
    print("Cube root = ", cube_root)
else:
    print("Please enter a number between 0 and 999.")
```

3. Read the birth date and salary in rupees of employees. Perform data transformation for birthdate

to age and also salary which is in rupees to salary in dollars using functions.

Ans :

```
from datetime import datetime
def calculate_age(birthdate):
    today = datetime.now()
    birthdate = datetime.strptime(birthdate, "%Y-%m-%d")
    return today.year - birthdate.year - ((today.month, today.day) <
(birthdate.month, birthdate.day))
def salary_in_dollars(salary):
    return salary /82.5

birthdate = input("Enter birthdate (YYYY-MM-DD): ")
salary = int(input("Enter salary in rupees: "))
```

```
age = calculate_age(birthdate)
salary_usd = salary_in_dollars(salary)

print("Age:", age)
print("Salary in USD: ", round(salary_usd, 2))
```

4. Print the reverse number of a given number

Ans :

```
num = input("Enter a number: ")
print("Reverse number is:", num[::-1])
```

Lab Assignment:

1. To accept students five courses marks and compute his/her result. Student is passing if he/she

scores marks equal to and above 40 in each course.

If student scores aggregate greater than 75 percentage, then the grade is distinction.

If aggregate is greater than or equal to 60 and less than 75 then the grade if first division.

If aggregate is greater than or equal 50 and less than 60, then the grade is second division.

If aggregate is greater than or equal 40 and less than 50, then the grade is third division.

Ans :

```
marks = [int(input(f"Enter marks {i}: ")) for i in range(1, 6)]
if min(marks) < 40:
    print("Fail")
else:
    per = sum(marks) / 500 * 100
    print("Percentage:", per)
    if per > 75:
        print("Distinction")
    elif per >= 60:
        print("First Division")
    elif per >= 50:
        print("Second Division")
    else:
        print("Third Division")
```

2. Solve the Fibonacci sequence using recursive function in Python

Ans :
```
def fib(n):
    if n <= 1:
        return n
    return fib(n-1) + fib(n-2)
    n = int(input("Enter number: "))
print(fib(n))
```

3. Write a Python program to print different patterns.

Ans :
```
n = int(input("Enter number: "))
for i in range(1, n+1):
    print("*" * i)

n = int(input("Enter number: "))
for i in range(n, 0, -1):
    print("*" * i)
```

Self Study Assignment:

Perform all statistical analysis (Average, Max, Min, Sum, on emplyoee data

Ans :

```
n = int(input("Enter number of employees: "))
emp = []
for i in range(n):
    emp.append(int(input("Enter salary: ")))

print("Sum =", sum(emp))
print("Average =", sum(emp)/n)
print("Max =", max(emp))
print("Min =", min(emp))
```

practical no.2

Practice Lab Assignment:2

1. Perform all List Operation, Dictionary Operations.

```
student={"name":"yash chaudhari","age":18,"grades":[85,90,98]}
print("Name:",student["name"])
```

Name: yash chaudhari

```
student["PRN"] = 202501080014
print("After adding PRN:",student)
```

After adding PRN: {'name': 'yash chaudhari', 'age': 18, 'grades': [85, 90, 98], 'PRN': 202501080014}

```
student["age"] = 19
print("After updating age:",student)
```

After updating age: {'name': 'yash chaudhari', 'age': 19, 'grades': [85, 90, 98], 'PRN': 202501080014}

```
student.pop("grades")
```

[85, 90, 98]

```
print("After removing grades:",student)
```

After removing grades: {'name': 'yash chaudhari', 'age': 19, 'PRN': 202501080014}

```
keys = student.keys()
print("Keys:", keys)
```

Keys: dict_keys(['name', 'age', 'PRN'])

```
values = student.values()

print("Values:",values)
```

Values: dict_values(['yash chaudhari', 19, 202501080014])

```
student_copy = student.copy()

print("Copy of dictionary:", student_copy)
```

Copy of dictionary: {'name': 'yash chaudhari', 'age': 19, 'PRN': 202501080014}

```
student.clear()

print("After clear:", student)
```

After clear: {}

LAB ASSIGNMENT 2:

1. **Select the number from the entered list and find its position in Python (use Linear Search).**

Code:

```
lst = []
n = int(input("Enter number of elements: "))
for i in range(n):
    lst.append(int(input("Enter number: ")))
x = int(input("Enter number to search: "))
for i in range(n):
    if lst[i] == x:
        print("Position:", i+1)
        break
else:
    print("Number not found")
```

2. Choose cricket team of eleven players find the captain of the team (consider tallest person as a captain) using dictionary.

Code:

```
team = {}
for i in range(11):
    name = input("Enter player name: ")
    height = int(input("Enter height: "))
```

```
    team[name] = height
max_height = 0
captain = ""
for name in team:
    if team[name] > max_height:
        max_height = team[name]
        captain = name
print("Captain of the team is:", captain)
```

SELF STUDY ASSIGNMENT 2:

1. Consider the student subjects marks are stored in the list. Find the total marks and percentage of students.

Code:

```
marks = []
n = int(input("Enter number of subjects: "))
for i in range(n):
    m = int(input("Enter marks: "))
    marks.append(m)
total = sum(marks)
percentage = total / n
print("Total Marks:", total)
print("Percentage:", percentage, "%")
```

2. Perform all tuple operations.

Code:

```
t = (10, 20, 30, 40, 50)
print("Tuple:", t)
Tuple: (10, 20, 30, 40, 50)
print("Length:", len(t))
Length: 5
print("First element:", t[0])
```

```
First element: 10

print("Slice (1 to 3):", t[1:4])

Slice (1 to 3): (20, 30, 40)

print("Count of 20:", t.count(20))

Count of 20: 1

print("Index of 30:" t.index(30))

SyntaxError: invalid syntax. Perhaps you forgot a comma?

print("Index of 30:", t.index(30))

Index of 30: 2

t2 = (60, 70)

print("After adding:", t + t2)

After adding: (10, 20, 30, 40, 50, 60, 70)
```